

Agrobase SaaS Pricing, Architecture & Infrastructure Plan

Prepared for: MobiPay AgroSys Limited — Founder & Engineering Review **Scope:** Multi-tenant SaaS pricing model, terms of service framework, full architecture (data, workflow, payment flows), and cloud infrastructure cost comparison **Date:** June 2026

1. Pricing model — what we charge, and on what basis

The core principle: **different actor types create value differently, so they should be charged differently.** A flat per-tenant fee leaves money on the table from high-volume actors (exporters, banks) and overcharges low-volume ones (small cooperatives).

1.1 Pricing by actor type

Actor type	Pricing basis	Billing trigger
NGO / development programme	Recurring subscription (per module bundle)	Monthly or annual, billed in advance
Cooperative	Recurring subscription (tiered by member count)	Monthly, billed in advance
Agribusiness (e.g. Nile Breweries)	Subscription + per-farmer activity fee	Monthly base + usage-based add-on, billed in arrears for usage
Trader / exporter	Transaction fee on marketplace activity + optional subscription for premium access	Per transaction, deducted at settlement
Input supplier	Listing fee + transaction commission	Per listing (flat) + per completed sale (%)
Processing company	Subscription (traceability/intake module) + per-batch fee	Monthly + per processing batch logged
Microfinance institution / bank	Per-query fee for credit score API + origination fee on loans facilitated	Per API call (metered) + % of loan value disbursed
Individual farmer (direct)	Freemium core + paid add-ons (advisory, premium marketplace access)	Monthly micro-subscription via mobile money

1.2 Marketplace — how we earn without owning inventory

This is the piece worth being precise about, since it's structurally different from the subscription streams.

MobiPay never takes title to goods. We are not a buyer or reseller — we are the platform that connects a seller (farmer, cooperative) to a buyer (trader, exporter, processor) and **we get paid at the moment money changes hands, not the moment goods change hands.**

The revenue components on every marketplace transaction:

1. **Platform/transaction fee** — a percentage (recommend 1.5–3%) of the transaction value, deducted automatically at settlement. This is the core marketplace revenue line.
2. **Payment processing fee** — a smaller percentage (0.5–1%) covering the cost of moving money through mobile money or bank rails. This can be passed through near-cost or marked up slightly.
3. **Listing fee (optional)** — a flat fee for input suppliers or exporters to list products/sourcing requests prominently. Useful for B2B-heavy categories where transaction volume alone won't generate enough revenue.
4. **Premium placement / featured listing fee** — sellers or buyers pay extra for visibility (top of search, featured badge).
5. **Verified buyer/seller fee** — a one-time or annual fee for compliance-verified status (EUDR-ready, GAP-certified) — valuable because it lets a buyer filter by verified-compliant sellers only.

When do we actually get paid? At settlement — not at order placement, and not at delivery. The flow (see diagram above) is:

1. Buyer pays the full order value into an **escrow account** (held by MobiPay's payment partner or a regulated trust account — MobiPay does not custody funds directly without the appropriate licensing).
2. Seller delivers the goods.
3. Buyer confirms receipt (or funds auto-release after a defined grace period, e.g. 5–7 days, to prevent buyers withholding confirmation indefinitely).
4. The **settlement engine** deducts MobiPay's platform fee and payment processing fee from the escrowed amount.
5. The **net amount** is released to the seller via mobile money or bank transfer.

This is the same structural pattern used by Jumia, Alibaba's trade assurance, and most B2B agri-marketplaces — the platform is a paid intermediary at the payment layer, never a counterparty in the trade itself.

1.3 Indicative pricing — first cut

Stream	Suggested rate
NGO/programme subscription	\$500–\$2,000/month, tiered by module count and farmer volume
Cooperative subscription	\$50–\$200/month, tiered by member count

Stream	Suggested rate
Agribusiness subscription	\$2,000–\$8,000/month base + \$0.50–\$2 per farmer activity logged
Marketplace transaction fee	1.5–3% of transaction value
Payment processing fee	0.5–1% of transaction value
Listing fee	\$5–\$50 per listing, depending on category
Credit score API query	\$0.50–\$2.00 per query
Loan origination fee	1–3% of disbursed loan value
Verified compliance status fee	\$20–\$100/year per verified plot or batch

2. Terms and conditions — framework

This is not a substitute for legal drafting — MobiPay should have this reviewed by counsel in each operating country — but the commercial terms below are the structure to put in front of a lawyer.

2.1 Subscription terms (NGOs, cooperatives, agribusiness)

- **Billing cycle:** monthly or annual, in advance. Annual commitments should carry a discount (10–15%) to improve cash flow predictability.
- **Module changes:** tenants can upgrade modules immediately (prorated charge); downgrades take effect at the next billing cycle (no mid-cycle refunds).
- **Data ownership:** the tenant owns the data they input; MobiPay has a license to process it for service delivery, aggregated analytics, and (with consent) credit scoring/third-party sharing.
- **Termination:** either party may terminate with 30 days' notice; tenant data is exportable for 90 days post-termination, then deleted per data protection law.
- **SLA:** uptime commitment (recommend 99.5% for paid tiers), defined support response times by tier.

2.2 Marketplace terms (traders, exporters, input suppliers)

- **Escrow terms:** funds are held until delivery confirmation or auto-release after the grace period; disputes pause auto-release pending resolution.
- **Fee disclosure:** platform fee and payment processing fee disclosed upfront on every transaction — no surprise deductions at settlement.
- **Dispute resolution:** a defined process (claim window, evidence submission, MobiPay mediation) before funds are released or returned.

- **Liability:** MobiPay is a facilitator, not a party to the underlying sale contract — quality, delivery, and compliance disputes are between buyer and seller, with MobiPay providing the traceability/evidence record but not guaranteeing the goods.
- **Compliance representations:** sellers attest to the accuracy of compliance data (EUDR, GAP status); MobiPay is not liable for fraudulent attestations but should have a verification/audit mechanism for high-value or flagged transactions.

2.3 Financial institution terms (banks, MFIs)

- **Data access scope:** explicit, farmer-consented scope of data shared (credit score output, not raw transaction history, unless separately agreed).
- **Query-based billing:** metered API usage, invoiced monthly in arrears, with a defined dispute window for contested queries.
- **No guarantee of creditworthiness:** MobiPay provides a score based on defined inputs; lending decisions and associated risk remain with the financial institution.

2.4 Data protection commitments (applies to all tenants)

- Compliance with Uganda's Data Protection and Privacy Act, Ghana's Data Protection Act, and Kenya's Data Protection Act, as applicable per tenant's operating country.
 - Farmer consent is explicit, revocable, and tracked per data-sharing purpose (e.g. consent to share with Bank X is separate from consent to receive SMS advisories).
 - Data breach notification commitments per local law (typically 72 hours).
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3. Full multi-tenant architecture — data, workflow, and payment flows

(see the two diagrams above for the visual reference)

3.1 Data flow

Field data originates at the mobile app layer (farmer registration, cultivation logs, harvest records, marketplace listings), captured offline-first and synced when connectivity is available. Every request passes through a **tenant-scoped API gateway**, which validates both the tenant's identity and which modules that tenant is entitled to before any request reaches a service — this is the enforcement point for the SaaS model, not just a security gate.

Validated changes are published to a shared **event bus**, every event tagged with `(tenant_id)`. Downstream services (compliance engine, credit scoring, marketplace/settlement) consume events independently, each writing back to a **tenant-isolated data layer** using row-level security keyed by `(tenant_id)`. MobiPay's super admin sits above all tenants with a **federated view** — able to query and monitor across every country and tenant without each tenant being able to see another's data.

3.2 Workflow — onboarding a new tenant

This is the operational workflow that makes "just create a new tenant" actually true in practice:

1. Super admin creates a new tenant record (name, country, tenant type — NGO/cooperative/exporter/etc.).
2. Super admin assigns a **module bundle** from the catalog (e.g. Training + Survey for an NGO; Marketplace + Compliance for an exporter).
3. The billing engine auto-generates a subscription plan matching the assigned modules and tenant tier.
4. Tenant admin account is provisioned; tenant admin can then onboard their own users (farmers, agents, staff) within their tenant scope.
5. First invoice is generated per the agreed billing cycle; module access activates on payment confirmation (or on a defined trial period, if offered).
6. Ongoing: module changes, billing changes, and usage are all visible to both the tenant admin (their own data) and the super admin (cross-tenant view).

3.3 Payment flow — by revenue type

- **Subscription revenue:** charged automatically per billing cycle via mobile money, card, or bank transfer integrated into the billing engine; failed payments trigger a grace period then module suspension.
- **Marketplace revenue:** captured at settlement, as described in Section 1.2 — MobiPay's fee is deducted from the escrowed transaction amount, never invoiced separately to the seller.
- **Credit score API revenue:** metered per query, invoiced monthly in arrears to the financial institution tenant.
- **Loan origination revenue:** deducted from the disbursed loan amount at the point of disbursement, per the agreement with the partner MFI/bank.

4. Infrastructure cost — monthly CapEx/OpEx estimate

These are **indicative, planning-stage estimates** for a platform at early-to-mid scale (roughly 300k farmers, growing into Ghana/Kenya, with marketplace and compliance workloads live). Actual costs should be validated with a proof-of-concept load test before committing to a provider.

4.1 Estimated monthly cloud spend by growth stage

Stage	Description	Estimated monthly cost (USD)
MVP / Phase 1	Core services, single region, Uganda only, low traffic	\$800-\$1,500

Stage	Description	Estimated monthly cost (USD)
Growth / Phase 3-4	Multi-service, compliance engine, satellite processing, marketplace live	\$3,000-\$7,000
Multi-country scale	3 countries live, full traceability + blockchain anchoring, high mobile sync volume	\$8,000-\$18,000

These ranges include compute (Kubernetes cluster), managed database, object storage, event streaming, CDN, monitoring/logging, and reasonable data transfer — they exclude third-party API costs (satellite imagery provider, SMS gateway, mobile money transaction fees) which are typically usage-based and billed separately.

4.2 AWS vs GCP vs Azure — comparison for this workload

Factor	AWS	GCP	Azure
Managed Kubernetes	EKS — mature, widest ecosystem	GKE — generally regarded as the most polished/easiest to operate	AKS — solid, strong if already in Microsoft ecosystem
Managed PostgreSQL + PostGIS	RDS for PostgreSQL — full PostGIS support	Cloud SQL for PostgreSQL — full PostGIS support	Azure Database for PostgreSQL — full PostGIS support
Geospatial / satellite tooling	Strong via partner ecosystem (no native Earth Observation product)	Google Earth Engine — significant advantage for satellite/deforestation analysis, directly relevant to EUDR monitoring	Azure Planetary Computer — also strong geospatial offering, newer
Event streaming (Kafka)	MSK (managed Kafka)	Confluent Cloud or self-managed on GKE	Event Hubs (Kafka-compatible) or self-managed
Africa region presence	Cape Town region (closest to East Africa)	No dedicated Africa region — routes via Europe	No dedicated Africa region in East Africa — routes via Europe/UAE
Pricing transparency	Complex but well-documented, many cost calculators	Generally considered most cost-predictable, sustained-use discounts automatic	Comparable to AWS in complexity
Mobile money / fintech partner	Largest ecosystem of African fintech	Smaller footprint in African fintech specifically	Smaller footprint in African fintech

Factor	AWS	GCP	Azure
ecosystem	integrations and case studies		specifically
Best fit for this platform	Strongest overall ecosystem maturity + closest regional presence (Cape Town)	Strongest fit if satellite/EUDR monitoring via Earth Engine is a priority — meaningfully reduces build effort there	Reasonable if there's an existing Microsoft/Azure relationship, otherwise no compelling edge for this specific workload

Recommendation: AWS is the safer default given its Cape Town region (lowest latency for East African users) and the largest base of African fintech/mobile-money integration patterns to draw from. However, if the satellite/EUDR deforestation monitoring workload turns out to be a major engineering lift, it is worth pricing out a **hybrid approach** — core platform on AWS, satellite/Earth Observation processing on GCP via Google Earth Engine, which has no real equivalent in AWS or Azure's native tooling. This hybrid pattern is common in agri-tech specifically for this reason.

4.3 CapEx vs OpEx framing

Cloud infrastructure of this kind is structurally **OpEx, not CapEx** — there is no large upfront infrastructure purchase. The "capital" investment is primarily:

- **Engineering build cost** (the team and time to build the platform — this is the real capital outlay, not the cloud bill)
- **Initial cloud spend during the build phase** (Phase 1-2, before any tenants are paying) — budget \$800-\$3,000/month during this period, before marketplace and multi-country workloads add cost
- **Third-party licensing/API setup costs** (satellite imagery provider contracts, blockchain infrastructure if not self-hosted, payment gateway integration fees) — typically one-time setup fees plus usage-based ongoing cost

The cloud bill becomes self-funding once subscription and transaction revenue covers it — at the Phase 3-4 stage (\$3,000-\$7,000/month), even a modest 10-15 paying enterprise tenants at the subscription rates in Section 1.3 covers infrastructure cost with margin to spare.

5. Summary

The pricing model charges each actor type on the basis that matches how they extract value — subscription for steady platform access, transaction fees for marketplace activity, metered fees for credit scoring API access. The marketplace specifically earns through an escrow-and-settlement model: MobiPay is paid at the moment of settlement, taking a

platform fee and processing fee out of funds already collected from the buyer, without ever holding inventory or taking trade risk.

The multi-tenant architecture enforces this pricing model technically, not just commercially — module entitlement and billing are the same system, so assigning a tenant a module bundle is what generates their invoice. Infrastructure cost scales with usage (OpEx), starting under \$1,500/month and growing into the \$8,000–\$18,000/month range only once the platform is generating revenue at multi-country scale to cover it.

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